UbiComp ISWC 2026

香 港 大 學 THE UNIVERSITY OF HONG KONG

HKU AIoT LAB 香港大學人工智能物联网實驗室

A! Aalto University School of Electrical Engineering

Ambient Intelligence

Tutorial

Towards Open-Sourced Datasets, Benchmarks, and Platforms for Multi-Modal Sensing



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Computing as Science

SCIENCE | INTERNET | AI

1946 | 1965 | 1980 | 1995 | 2010 | 2020








First Computer | High Perf. Computing | Personal Computer | Internet | IoT & CPS | **AIoT** Connectivity & Intelligence

Human-Machine | Machine-Machine | Things-Things | Agents-Agents

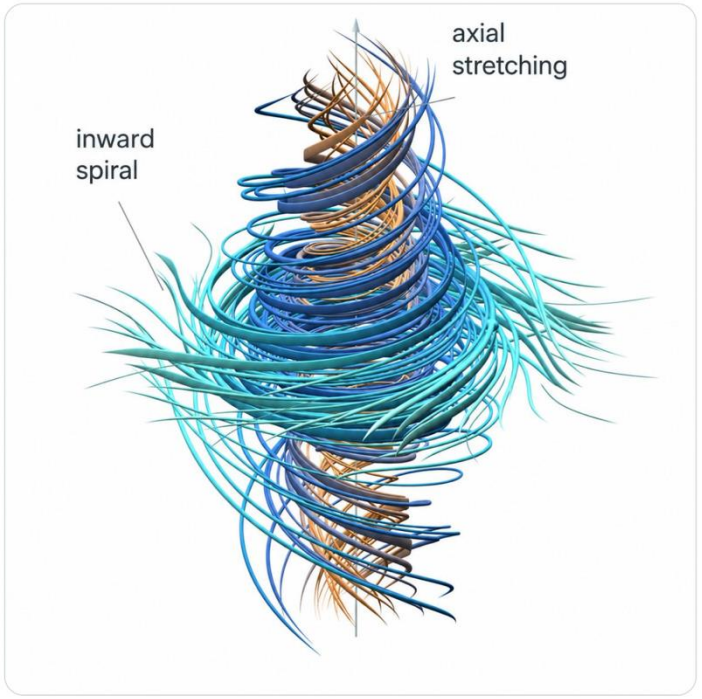
AI 2026: A Giant in Thought, a Dwarf in Action

OpenAI ✓
@OpenAI

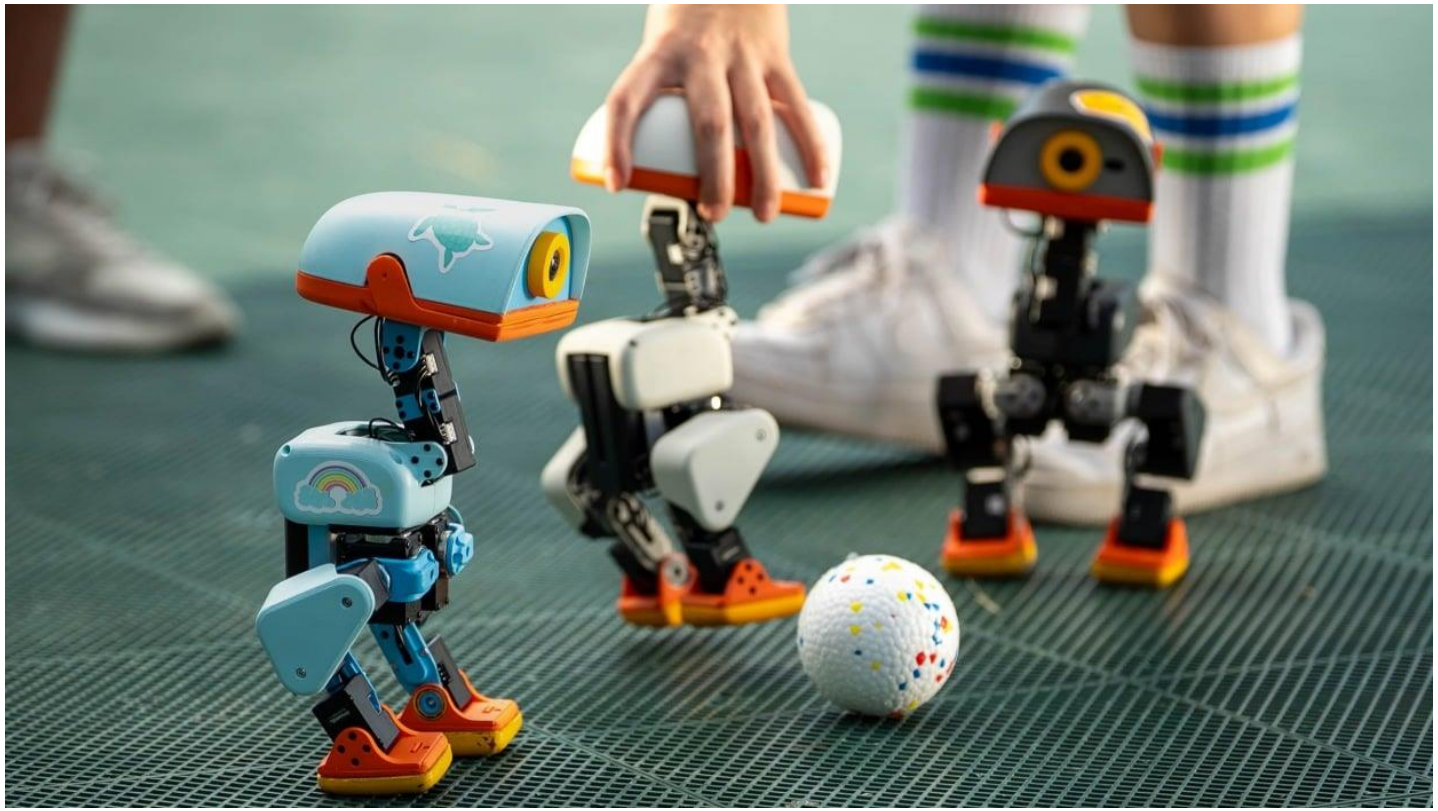
We're sharing a solution to the Navier-Stokes Millennium Prize Problem, one of the deepest problems at the frontier of mathematics.

The proof was produced by a group of agents, using an OpenAI next-generation model significantly more capable than GPT-6 Astra.

The problem concerns whether the description of smooth three-dimensional fluid motion modeled by the Navier-Stokes equations can break down. It has remained unresolved for roughly 90 years.



1:20 AM · Sep 9, 2026 · **46.9M** Views

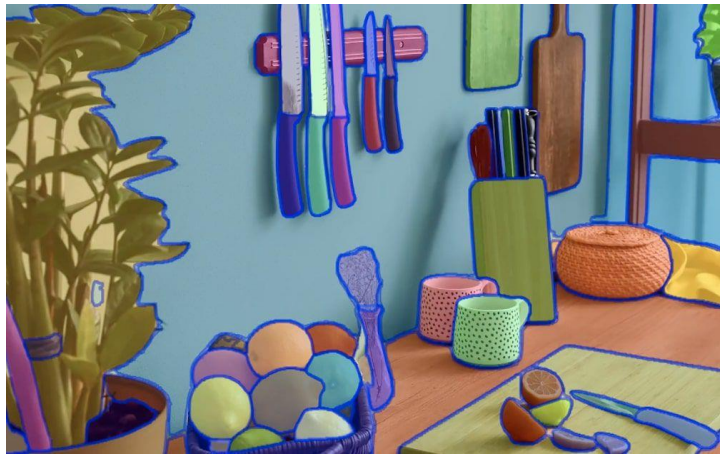


Today's AI has BIG Success













Cannot “see” in the dark or through the wall;
Neither with privacy.

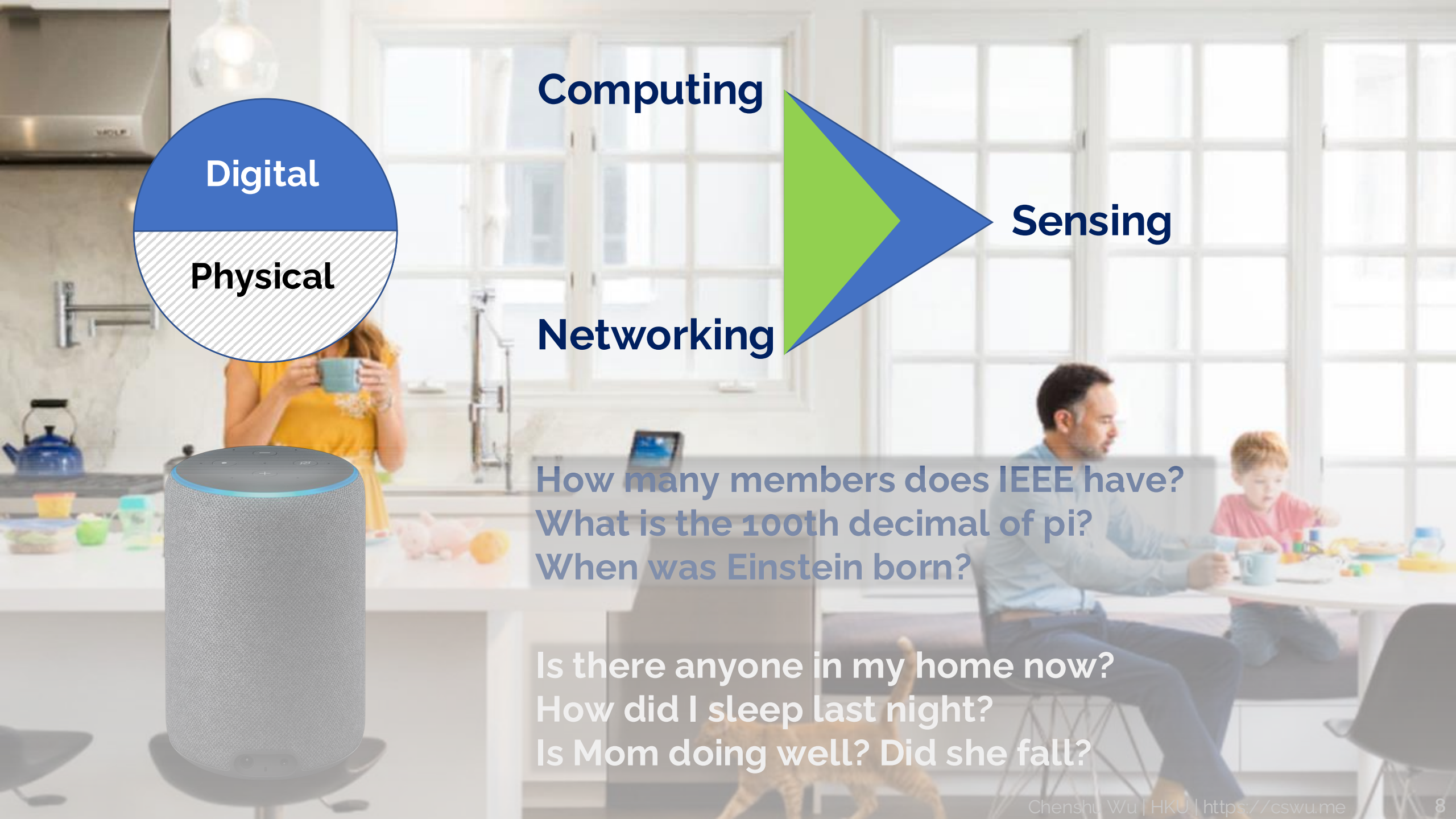
Today's AI has BIG Problem

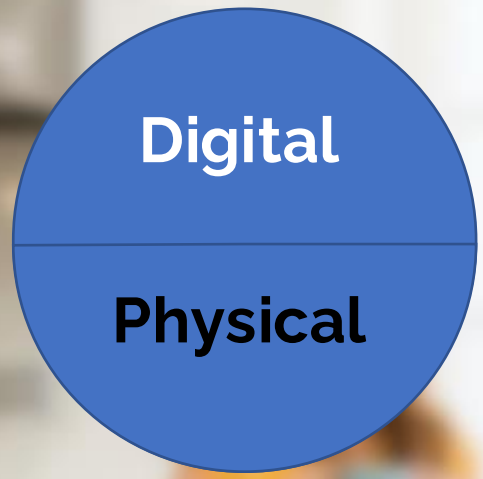


GET SMALL, GET PHYSICAL

TOO BIG

VIRTUAL


Digital
Physical
Computing
Networking
Sensing
How many members does IEEE have?
What is the 100th decimal of pi?
When was Einstein born?
Is there anyone in my home now?
How did I sleep last night?
Is Mom doing well? Did she fall?



Computing

Networking

Sensing



Sensing AI

Equip connected devices with sensing AI to bridge the cyber and physical spaces

Sensing AI: Machine Intelligence beyond Vision and Language



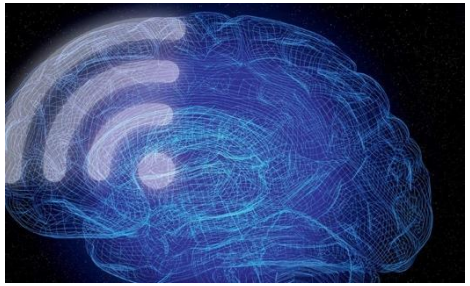
Computer Vision

deals with how computers understand visual signals



Natural Language Processing

enables machines to understand natural language/text data



Sensing AI

enables **IoT** to sense the physical world with **ambient signals**

Photo courtesy: *

*: Liu, K., & Wang, B. (2019). Wireless AI: Wireless Sensing, Positioning, IoT, and Communications. Cambridge: Cambridge University Press.

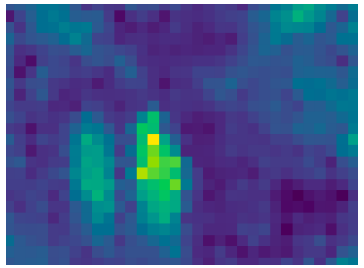
Sensing AI Beyond Vision

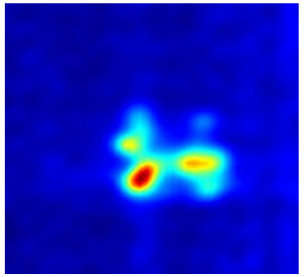

Visual Camera

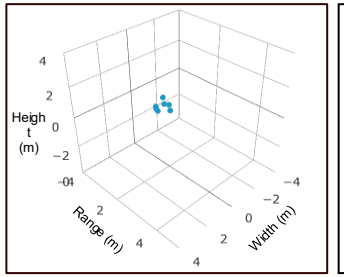

IR Camera



Thermal Camera


Thermal Array

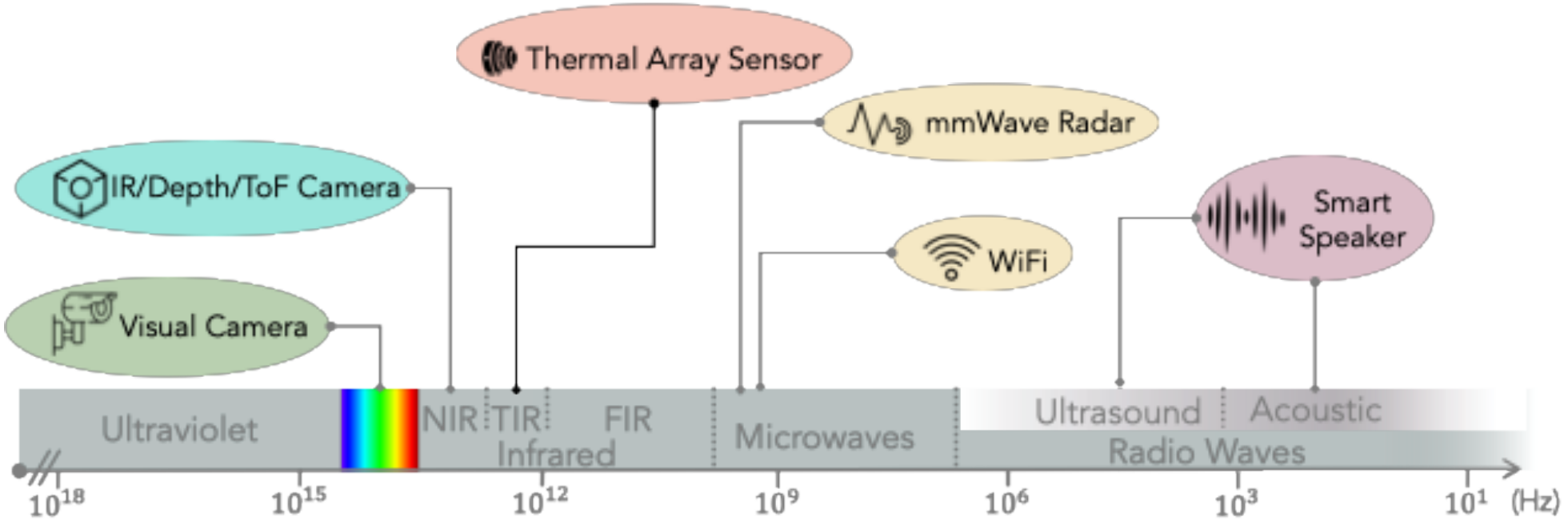

Vayyar mmWave



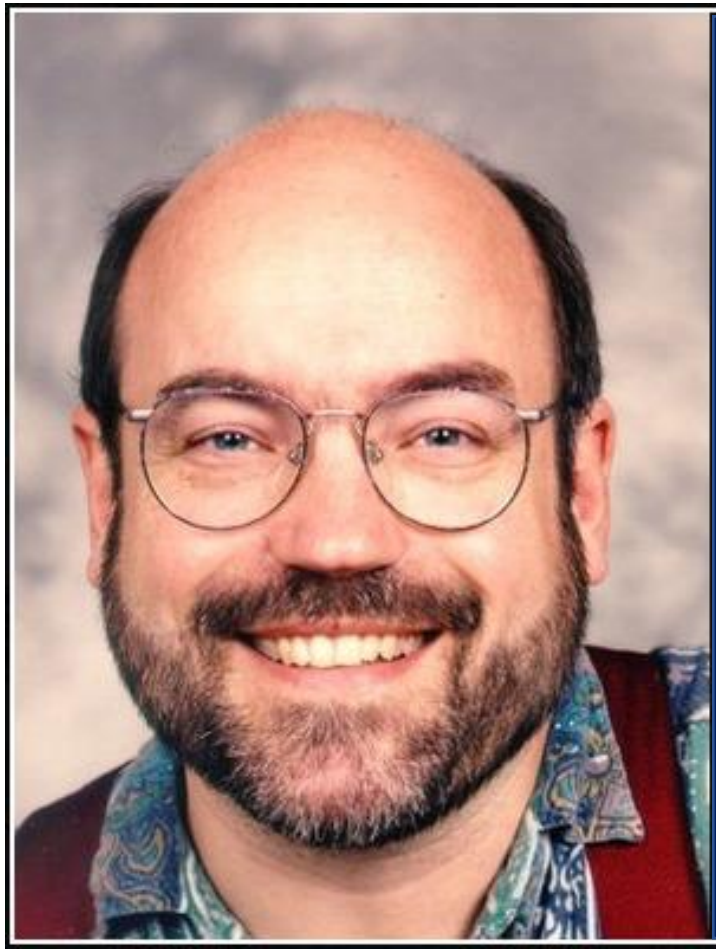
TI mmWave


WiFi & Acoustic

Image Resolution ⟷ **Privacy Protection**



Wearables vs “Invisibles”?



The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it.

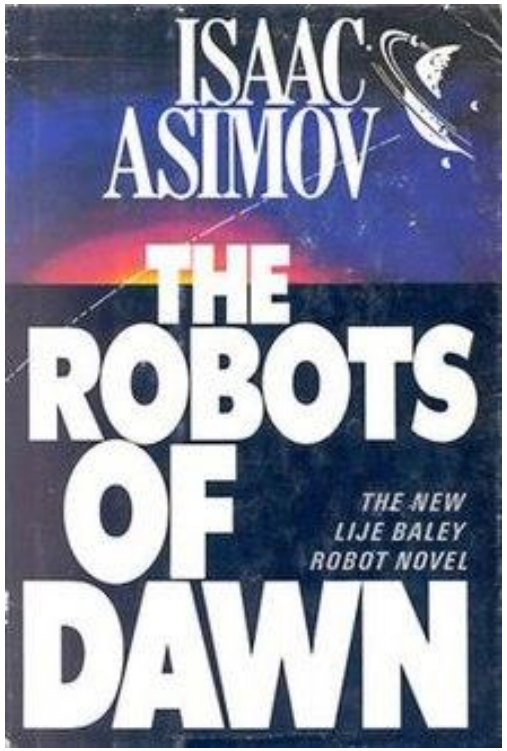
-- Mark Weiser
(The Father of Ubiquitous Computing)



The Computer for the 21st Century

Specialized elements of hardware and software, connected by wires, radio waves and infrared, will be so ubiquitous that no one will notice their presence

(Scientific American, 1991)



"You lift your arm whenever you want something, and the robots always know what it is. How did this one know you wanted tea served?"
"Every time I lift my arm, it distorts a small EM field that is maintained continuously across the room. Slightly different positions of my hand and fingers produce different distortions, and my robots can interpret these distortions as orders."

-- Isaac Asimov, "The Robots of Dawn", 1983

Wireless Sensing AI

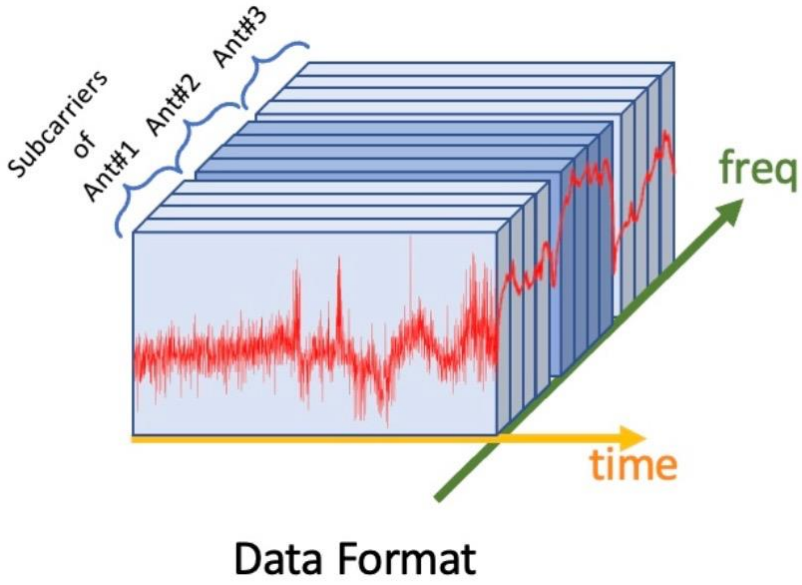
WiFi Can Do More!

- Motion Detection
- Sleep Monitoring
- Fall Detection
- Gait Recognition
- Gesture Control
- Wellbeing Monitoring
- Activity Monitoring
- Location Tracking
- Many More…



Deep Wireless Sensing (DWS)

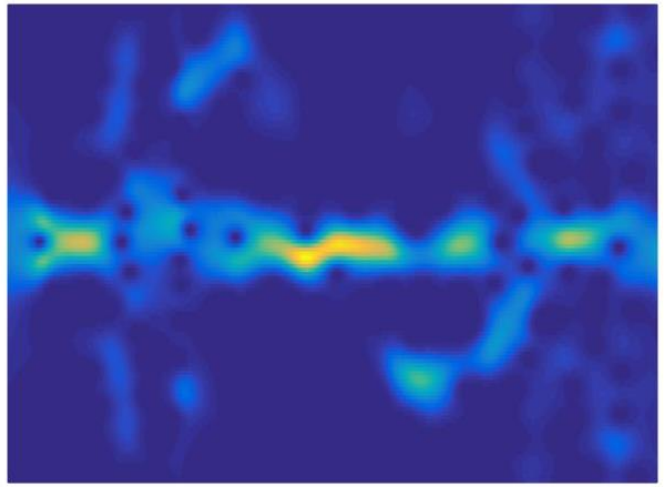
- **Non-visual**
  - Contain physical and geometric connotations in time, space, and frequency domains all non-visually intelligible (like images to human eyes)
  - (They can certainly be visualized)
  - Image data are visual, of course
- **Complex**
  - Complex-valued tensors with amplitude and phase
  - Visual data are real-valued
- **High-dimensional**
  - time, subcarrier, antennas, transceivers
  - Visual data are usually 2D (image) or 3D (video)
- **Difficult to collect & label, and learn**



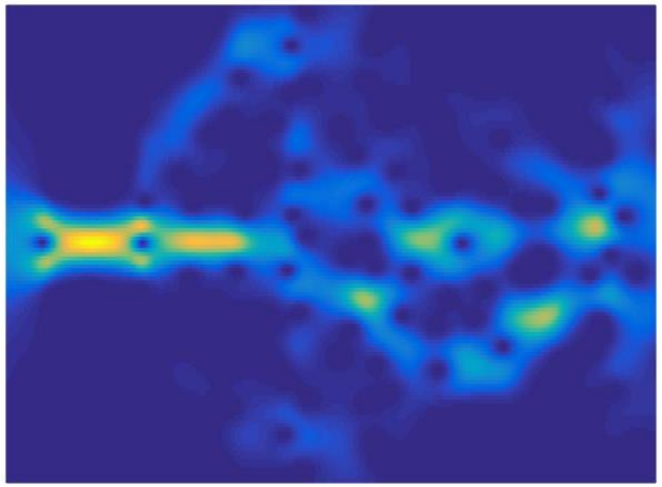
Data Format

Radio Frequency (RF) Datasets

- **More difficult to collect**
  - Much more complex setups
  - Less reliable platforms for data collection
    - Not always capture valid data
    - Not all captured data are useful/meaningful
  - Depends on many environmental factors
    - Users, device placements, user locations, user orientations, places/rooms, environmental factors like furniture, wireless configs, device heterogeneity…



CSI in Setting 1



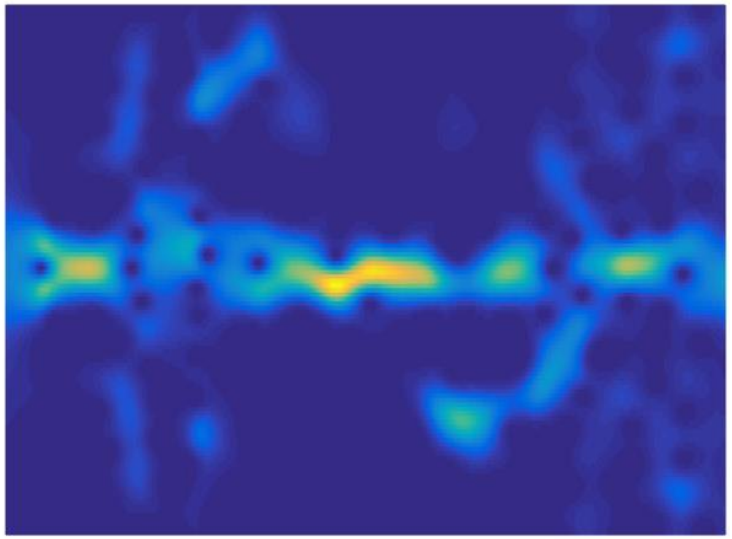
CSI in Setting 2

RF Datasets

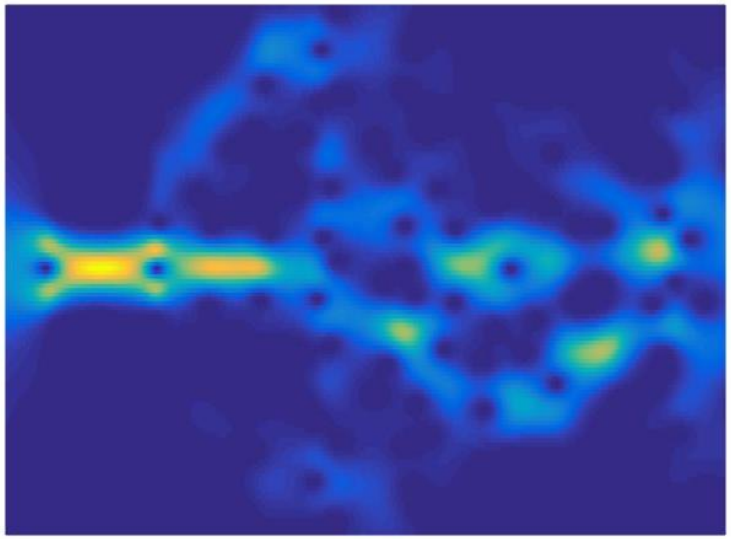
- **More difficult to label**
  - Cannot be labelled OFFLINE (unlike images!)



What are these?



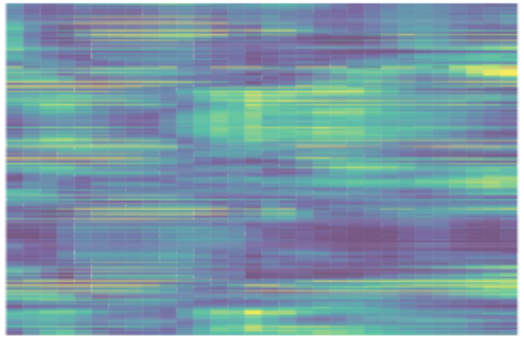
What is this?



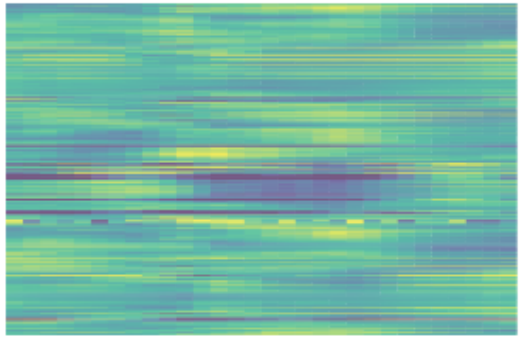
And this??

RF Datasets

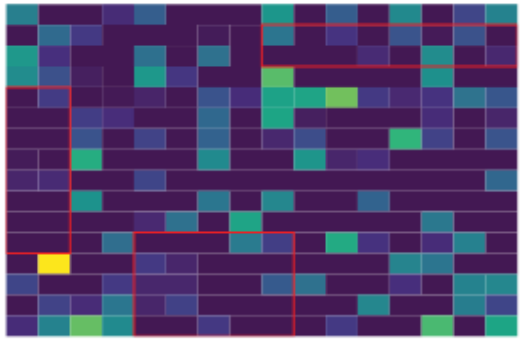
- Lack of large-scale datasets
  - **Widar3** dataset: 17 participants performing 22 gestures from 5 orientations towards one transmitter, standing at 5 different locations within the coverage of six receivers in 3 environments.
  - **30K** vs. **14M** (ImageNet)
- Lack of high-quality, easy-to-use datasets



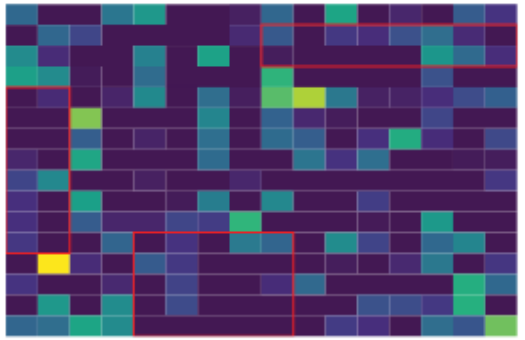
(a) Wi-Fi CSI in environment 1.



(b) Wi-Fi CSI in environment 2.



(c) Feature map of environment 1.



(d) Feature map of environment 2.

Wireless Sensing 2.0: SP-DL Co-Design

1) How to perform signal processing so that the processed data would produce better learning performance?
2) How to design neural networks so that they will better fit the unique wireless data (that are very different from visual data)?

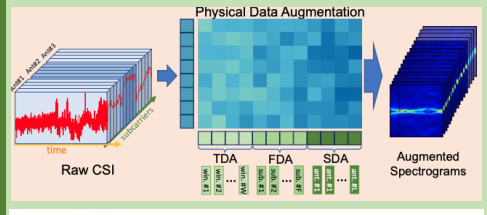


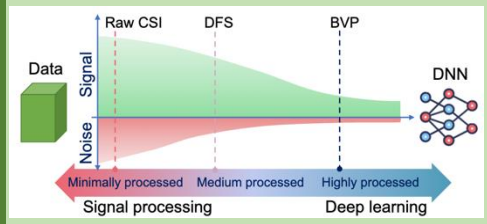
model-guided, data-driven design?

Our Attempts for Wireless Sensing 2.0

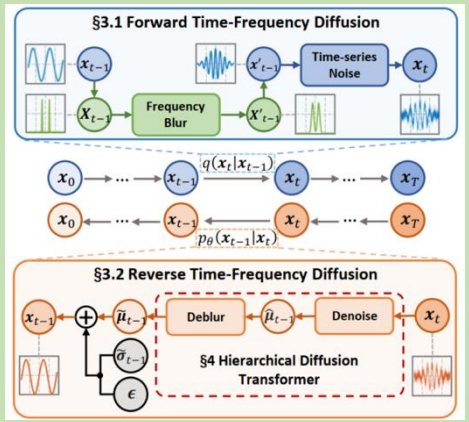
Datasets

RF Data Augmentation & Generation



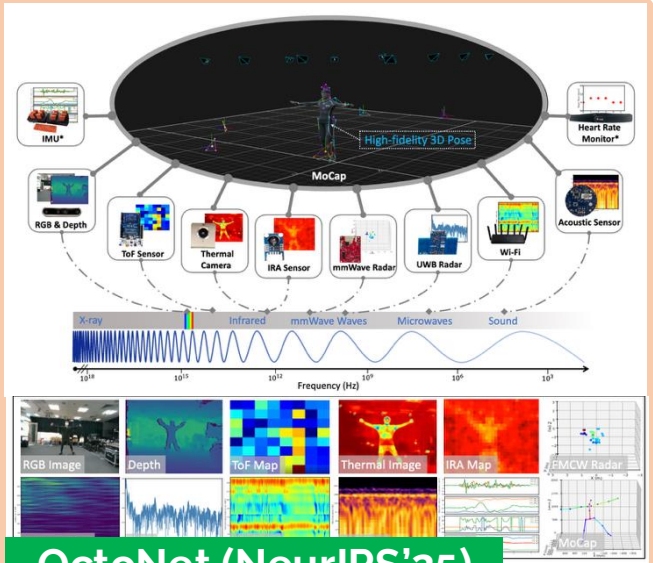


RFBoost (UbiComp'24)



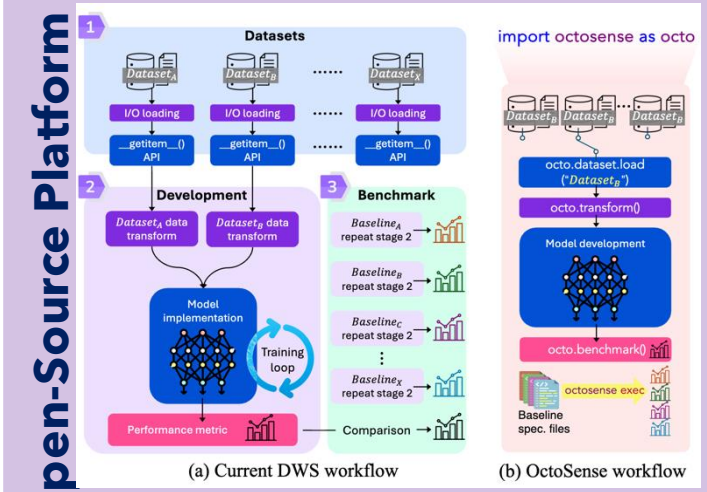
RF-Diffusion (MobiCom'24)

Large-Scale Open Datasets



OctoNet (NeurIPS'25)

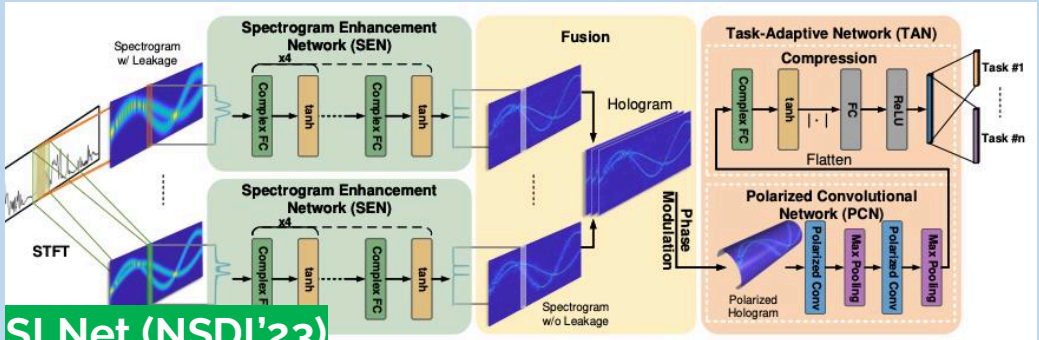
Open-Source Platform



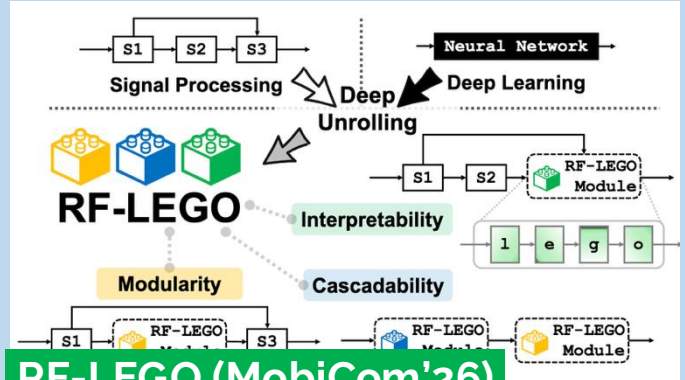
OctoSense (SenSys'27)

Models

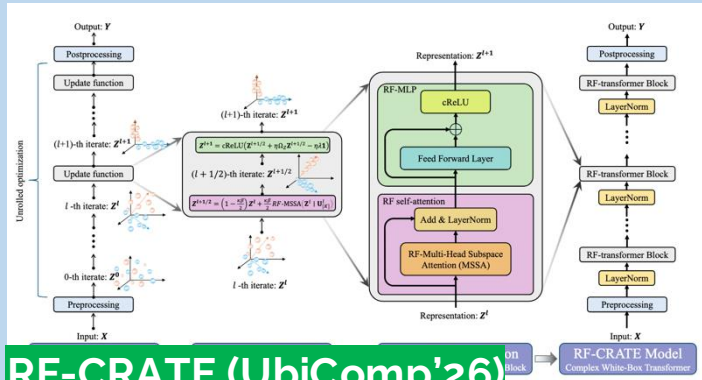
RF Neural Network Designs



SLNet (NSDI'23)



RF-LEGO (MobiCom'26)



RF-CRATE (UbiComp'26)

HOW, but not WHY

The Premise

Multi-modal architectures inherently outperform single-modal baselines.

The Objective

A starting point for HOW: Make datasets **available**, **usable**, and **learnable**.

> *"Now this is not the end. It is not even the beginning of the end. But it is, perhaps, the end of the beginning."*
>
> **Multi-modal sensing @2026**

Content

00 **Opening**

Motivation, scope, learning goals, and tutorial format.

01 **Module I · Datasets and benchmarks**

OctoNet, RF-Behavior, OctoSense

02 **Module II · Models and learning paradigms**

RF-CRATE, TAP3D, RF-Lego

03 **Guided hands-on**

Workflows for OctoNet and RF-Behavior: streaming download, modality loading, and visualization.

04 **Closing**

Open research questions and discussion.